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# Monsoon

A **monsoon** (/mɒnˈsuːn/) is traditionally a seasonal reversing wind accompanied by corresponding changes in precipitation,[1] but now used to describe seasonal changes in atmospheric circulation and precipitation associated with annual latitudinal oscillation of the Intertropical Convergence Zone, specifically between its limits to the north and south of the equator.



Advancing monsoon clouds and showers in Aralvaimozhy, near Nagercoil, Tamil Nadu, India

Usually, the term monsoon is used to refer to the rainy phase of a seasonally changing pattern, although technically there is a dry phase. The term is also used to describe locally heavy but short-term rains.[2][3]

The major monsoon systems of the world consist of the West African, South Asian–Australian, North American, and South American monsoons.



Monsoon clouds arriving at Port Blair, Andaman, India

The term was first used in English in British India and neighbouring countries, referring to big seasonal winds blowing from the Bay of Bengal and Arabian Sea in the southwest that brought heavy rainfall to the area.[4][5]

## Etymology

The etymology of the word monsoon is not wholly certain.[6] The English *monsoon* came from Portuguese *monção*, ultimately from Arabic موسم (*mawsim*, "season"), "but partly via early modern Dutch *monson*".[7]



Monsoon clouds over Lucknow, Uttar Pradesh, India

## History

### Asian monsoon

Strengthening of the Asian monsoon has been linked to the uplift of the Tibetan Plateau after the collision of the Indian subcontinent and Asia around 50 million years ago.[8] Because of studies of records from the Arabian Sea and that of the wind-blown dust in the Loess Plateau of China, many geologists believe the monsoon first

became strong around 8 million years ago. More recently, studies of plant fossils in China and new long-duration sediment records from the South China Sea led to a timing of the monsoon beginning 15–20 million years ago and linked to early Tibetan uplift.[9] Testing of this hypothesis awaits deep ocean sampling by the Integrated Ocean Drilling Program.[10] The monsoon has varied significantly in strength since this time, largely linked to global climate change, especially the cycle of the Pleistocene ice ages.[11] A study of Asian monsoonal climate cycles from 123,200 to 121,210 years BP, during the Eemian interglacial, suggests that they had an average duration of around 64 years, with the minimum duration being around 50 years and the maximum approximately 80 years, similar to today.[12]

A study of marine plankton suggested that the South Asian Monsoon (SAM) strengthened around 5 million years ago. Then, during ice periods, the sea level fell and the Indonesian Seaway closed. When this happened, cold waters in the Pacific were impeded from flowing into the Indian Ocean. It is believed that the resulting increase in sea surface temperatures in the Indian Ocean increased the intensity of monsoons.[13] In 2018, a study of the SAM's variability over the past million years found that precipitation resulting from the monsoon was significantly reduced during glacial periods compared to interglacial periods like the present day.[14] The Indian Summer Monsoon (ISM) underwent several intensifications during the warming following the Last Glacial Maximum, specifically during the time intervals corresponding to 16,100–14,600 BP, 13,600–13,000 BP, and 12,400–10,400 BP as indicated by vegetation changes in the Tibetan Plateau displaying increases in humidity brought by an intensifying ISM.[15] Though the ISM was relatively weak for much of the Late Holocene, significant glacial accumulation in the Himalayas still occurred due to cold temperatures brought by westerlies from the west.[16]

During the Middle Miocene, the July ITCZ, the zone of rainfall maximum, migrated northwards, increasing precipitation over southern China during the East Asian Summer Monsoon (EASM) while making Indochina drier.[17] During the Late Miocene Global Cooling (LMCG), from 7.9 to 5.8 million years ago, the East Asian Winter Monsoon (EAWM) became stronger as the subarctic front shifted southwards.[18] An abrupt intensification of the EAWM occurred 5.5 million years ago.[19] The EAWM was still significantly weaker relative to today between 4.3 and 3.8 million years ago but abruptly became more intense around 3.8 million years ago[20] as crustal stretching widened the Tsushima Strait and enabled greater inflow of the warm Tsushima Current into the Sea of Japan.[21] Circa 3.0 million years ago, the EAWM became more stable, having previously been more variable and inconsistent, in addition to being enhanced further amidst a period of global cooling and sea level fall.[22] The EASM was weaker during cold intervals of glacial periods such as the Last Glacial Maximum (LGM) and stronger during interglacials and warm intervals of glacial periods.[23] Another EAWM intensification event occurred 2.6 million years ago, followed by yet another one around 1.0 million years ago.[19] During Dansgaard–Oeschger events, the EASM grew in strength, but it has been suggested to have decreased in strength during Heinrich events.[24] The EASM expanded its influence deeper into the interior of Asia as sea levels rose following the LGM;[25] it also underwent a period of intensification during the Middle Holocene, around 6,000 years ago, due to orbital forcing made more intense by the fact that the Sahara at the time was much more vegetated and emitted less dust.[26] This Middle Holocene interval of maximum EASM was associated with an expansion of temperate deciduous forest steppe and temperate mixed forest steppe in northern China.[27] By around 5,000 to 4,500 BP, the East Asian monsoon's strength began to wane, weakening from that point until the present day.[28] A particularly notable weakening took place ~3,000 BP.[29] The location of the EASM shifted multiple times over the course of the Holocene: